1. Objective
Monitor interior cabin climate and vehicle ride dynamics using ESP32-C3, and publish the data over MQTT for fleet or personal dashboard use.

2. Hardware

- MCU: ESP32-C3 dev board
- Climate sensor: DHT22 (temperature, humidity)
- IMU: MPU6050 (3-axis accelerometer)
- Power: 5V via micro-USB (from 12V automotive adapter)

Pin Mapping

Signal	ESP32-C3-Mini Pin
DHT22 VCC	3.3V
DHT22 GND	GND
DHT22 OUT	GPIO2
MPU6050 VCC	3.3V
MPU6050 GND	GND
MPU6050 SCL	GPIO8
MPU6050 SDA	GPIO9

3. Functional Requirements

- Read temperature and humidity from DHT22 every 2 seconds
- Read X, Y, Z acceleration from MPU6050 and convert to G-force
- Map axes: X = lateral (cornering), Y = longitudinal (accel/braking), Z = vertical (bumps)
- Connect to WiFi automatically ; reconnect if the connection drops
    - SSID : GopalaSutharsan
    - Password : 123987456
- Publish climate and motion data as a single JSON payload to MQTT every 2000ms
- Maintain configurable alert thresholds for temperature, humidity, and G-force

- Publish immediately to the alert topic when any threshold is exceeded, independent of the regular 2000ms cycle

4. MQTT

Periodic topic: `vehicle/cabin/telemetry`, every 2000ms:


```
{
  "temp_c": 24.5,
  "humidity_percent": 42,
  "lateral_g": 0.32,
  "longitudinal_g": -0.18,
  "vertical_g": 1.05
}
```


Alert topic: `vehicle/cabin/alert`, event-driven:


```
{
  "temp_c": 47.8,
  "humidity_percent": 42,
  "lateral_g": 0.85,
  "longitudinal_g": -0.18,
  "vertical_g": 2.4,
  "trigger": "vertical_g"
}
```


5. Expected Use

Installed in a vehicle, streaming cabin comfort and ride data to a dashboard for driver behavior review or ride comfort tracking, with immediate alerts on extreme cabin or dynamics events.